DARIO FUMAROLA

New York, NY • hello@dariofumarola.com • +1 (540) 466-2013 • dariofumarola.com

EXPERIENCE

Solutions Engineer, AI Prototyping — New York, NY
Amazon Web Services (AWS) — *Jun 2023 – Present*

- Owned 20+ enterprise AI engagements from C-suite discovery through production, serving as hands-on technical lead across architecture, implementation, and evaluation. Led teams of 5+ engineers across concurrent 1–4 week production builds for customers representing $300M in AWS ARR.
- Took pharmaceutical ML teams from manual model deployment to an LLM agent, automating packaging, GPU scheduling, provisioning, validation, observability, and recovery. Iterated on prompts, tools, and evals against real deployment requests and added human approval gates, cutting deployment time from weeks to hours; adopted across 10+ teams.
- Codified an evaluation-driven deployment framework spanning representative test sets, deterministic checks, model graders, human review, and production telemetry; shipped reusable templates adopted across AWS Prototyping, cutting repeat prototype-to-production time by 30%.
- Diagnosed and fixed inference bottlenecks in production Amazon Bedrock workloads using dynamic batching, request scheduling, streaming, and custom CUDA and Trainium NKI kernels; reduced time to first token by 35%, increased throughput 1.8×, and upstreamed serving optimizations into Bedrock.
- Partnered with an autonomous-driving customer's perception team to diagnose and solve a training-data bottleneck; built a diffusion-based camera-to-radar synthesis pipeline from archived footage, avoiding 50K+ miles of new vehicle data collection and improving downstream perception by 12%.

SELECTED RESEARCH & IP

Safe, Social Pre-training of LLM Agents | *Amazon Research Award* — *2026 – Present*

- Technical co-lead on a $500K Build on Trainium project; built the RL training and evaluation stack for LLM agents, including checkpoint evals across reasoning, social inference, and training dynamics, plus the Neuron/XLA and NKI infrastructure it runs on.

Elastic State Models: Bounded Adaptive Inference | *NeurIPS* — *2025*

- Designed an inference-time layer that lets a frozen streaming state-space model recover from hard states within a fixed worst-case compute budget, with no retraining; increased navigation success from 68% to 85%, halved collisions, and reduced torsion-chain violations by 64%.

Mood Manifolds: Runtime Safety & Behavioral Steering | *ICML* — *2025*

- Developed a runtime control layer that steers embodied agent behavior from live hazard signals to trade off task reward, safety, and exploration without retraining; a single steerable policy covered most of the reward–safety frontier spanned by 64 separately trained specialist policies.

Selective Replication for Sharded Vector Search | *Amazon Science* — *2024*

- Co-invented the patented selective-replication method behind Amazon S3 Vectors, serving tens of billions of vectors in production; reduced recall error by 28% at a fixed vector-access budget.

TECHNICAL

AI / LLM Systems	Agentic systems & tool use, evaluation harnesses, retrieval/RAG, LLM inference & serving, reinforcement learning, multimodal systems, vector search
ML & Infrastructure	CUDA, Trainium/NKI, Neuron/XLA, distributed training, Kubernetes, OpenSearch, Redis, Neptune
Software	Python, C++, TypeScript/JavaScript, FastAPI, React, CDK, Terraform

EDUCATION

Washington and Lee University | *B.S. Computer Science and Mathematics, Davis UWC Scholar* — *2019 – 2023*